AI Playbook and Toolkit for Public Health Departments

Business Analysis for AI-Enabled Public Health Systems

PMG 200

Business analysis translates public health needs into clear requirements, workflows, user stories, acceptance criteria, data needs, and implementation decisions. AI projects often fail when technical teams, vendors, and program staff do not share a precise understanding of the problem to be solved. This module gives informaticians and program leads a disciplined approach to defining AI-supported workflows before technology choices are finalized.

Level	applied/foundational
Primary track	Program Management Role-Based Track
Appears in tracks	operations-data-quality

Learning Objectives

- Explain the purpose of business analysis for ai-enabled public health systems in public health AI work.
- Identify the technical, operational, privacy, security, equity, and governance issues associated with the topic.
- Apply the topic to a real or realistic public health workflow, system, or implementation scenario.
- Recognize common failure modes that can affect safety, accuracy, trust, workflow fit, or sustainability.
- Identify practical guardrails that should be documented before implementation.
- Produce a AI business analysis package that can support readiness assessment, governance review, procurement, implementation, or monitoring.

Module Overview

Business analysis translates public health needs into clear requirements, workflows, user stories, acceptance criteria, data needs, and implementation decisions. AI projects often fail when technical teams, vendors, and program staff do not share a precise understanding of the problem to be solved. This module gives informaticians and program leads a disciplined approach to defining AI-supported workflows before technology choices are finalized.

Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Business Analysis for AI-Enabled Public Health Systems matters because AI systems are embedded in public health programs, data systems, and decision processes rather than operating in isolation. The topic affects whether AI outputs are accurate, explainable, safe, equitable, secure, and sustainable. Agencies that ignore this area may create tools that work in a demonstration but fail in actual practice.

The topic also affects accountability. Public health agencies must be able to explain how data are used, why a tool was approved, what evidence supports the workflow, who reviews outputs, and what happens when the system fails. These responsibilities become more important as AI tools move from experimentation into operational use.

Core Concepts

Business analysis clarifies the problem, workflow, users, requirements, constraints, and success criteria before technology decisions are made. AI projects often fail when teams start with a tool rather than a well-defined public health need.

A user story describes what a user needs to do and why. In AI work, user stories should include human review, data requirements, documentation, and limitations. A user story should not simply say that the user wants AI; it should describe the task the AI is supporting.

Acceptance criteria are testable conditions that determine whether a requirement has been met. For AI systems, acceptance criteria should include accuracy, workflow fit, usability, security, privacy, accessibility, and human oversight.

Public Health Example

A local health department may want AI to help classify community complaints. A business analyst would clarify whether the real problem is high complaint volume, inconsistent routing, delayed response, lack of category definitions, or insufficient staffing. The resulting requirements may call for classification support with human review rather than fully automated complaint resolution.

Topic Deep Dive

Business analysis begins with problem definition. Teams should document the public health objective, the operational pain point, the affected users, the current workflow, the decision being supported, and the consequences of error. A vague goal such as “use AI to improve efficiency” is not sufficient for implementation.

Requirements should be grounded in workflow. Analysts should define current-state and future-state processes, data inputs, outputs, decision points, exception handling, reporting needs, and approval steps. This helps prevent tools from being procured without a clear implementation pathway.

AI business analysis also requires boundary setting. Teams should define intended uses, prohibited uses, users, data permissions, human review points, and escalation triggers. These boundaries should become requirements, training expectations, and governance criteria.

Risks, Failure Modes, and Guardrails

A common failure mode is treating the topic as a technical detail rather than a public health control. When staff do not connect technical decisions to authority, workflow, equity, privacy, security, and accountability, the agency may adopt a tool that appears useful but cannot be sustained or defended.

Another risk is relying on vendor claims without agency-defined evidence. Vendors may provide demonstrations, dashboards, or summary metrics that do not match the agency workflow or data environment. A guardrail is to require documentation, test results, acceptance criteria, and agency-controlled review.

A third risk is failing to plan for change. Public health data, policies, systems, and emergencies change. Any technical approach must include monitoring, review, versioning, escalation, and retirement pathways so the agency can respond when assumptions no longer hold.

Application to AI-Supported Workflows

Generative AI, predictive analytics, RAG, automation, and agentic workflows all depend on the controls described in this module. The controls should be scaled to the risk of the use case. A tool that drafts internal training text may require lighter review than a tool that updates case records, prioritizes outreach, or supports public communication during an emergency.

The module should be applied during readiness assessment, procurement, implementation planning, pilot testing, and post-deployment monitoring. Learners should document how the topic affects data, systems, users, safeguards, and decision-making for the selected workflow.

Reflection Questions

Where does this topic appear in one current or proposed AI workflow in your agency?

What decisions, data sources, users, or partners would be affected if this area is poorly designed?

What evidence would governance reviewers need before approving the workflow?

Which safeguards should be required before pilot testing?

Who owns ongoing monitoring and review?

Practical Exercise

Create a AI business analysis package for one real or realistic public health AI workflow. The artifact should describe the use case, affected users, data sources, system dependencies, risks, guardrails, review points, and unresolved questions.

The exercise should produce a work product that could be used in a governance meeting, procurement review, implementation planning session, pilot evaluation, or monitoring review. Learners should document assumptions and identify which staff or partners need to be consulted.

Expected Artifact or Evidence

The expected artifact is a completed AI business analysis package. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Expected Assignment

- The expected artifact is a completed AI business analysis package. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Knowledge Check

- 1. What is the best reason to address business analysis for ai-enabled public health systems before deployment?
- 2. Which practice best supports responsible implementation?
- 3. Which statement best describes a common failure mode?
- 4. What should be included in the practical artifact for this module?
- 5. When should the issue be escalated for governance or leadership review?

References and Resources

- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP Top 10 for LLM Applications, when security or AI integrations are relevant
- Agency privacy, cybersecurity, data governance, procurement, and records management policies